

Lecture 21: Game Theory

*Instructor: Jonathan Pipping-Gamón**Author: JP*

21.1 Decisions Against Decisions

Many sports-analytics problems treat the world as something we estimate: a conversion probability, an expected-points value, an expected-goals probability, a player effect, or a similarity score. Game theory begins when the world is also choosing back.

The running football example is a third-and-1 at midfield. The offense can run or pass. The defense can sell out against the run or the pass. The payoff is the offense's probability of converting the first down:

	Run defense	Pass defense
Run	40	70
Pass	70	40

The defense's payoff is the probability of a stop, so the same game can be written as

	Run defense	Pass defense
Run	(40, 60)	(70, 30)
Pass	(70, 30)	(40, 60)

where each cell is (offense conversion probability, defense stop probability).

Definition 21.1 (Utility). *A utility function assigns a number to each possible outcome so that larger numbers mean more desirable outcomes for that player.*

Utility is just a way to say what a player gets out of an outcome. It is not always “yards” or “points.” It is whatever objective the decision maker is using. On this play, we use conversion probability because the example is about one yard. In another setting, the same framework could use expected points, win probability, injury risk, fatigue, or a weighted combination of several objectives.

Definition 21.2 (Best Response). *A best response is the action that gives a player the highest utility against a specified opponent action or strategy.*

For the football matrix, the pure best responses are:

Question	Condition	Best response
Offense	Defense chooses Run defense	Pass (70 > 40)
Offense	Defense chooses Pass defense	Run (70 > 40)
Defense	Offense chooses Run	Run defense (60 > 30)
Defense	Offense chooses Pass	Pass defense (60 > 30)

This is the central habit of game theory: do not ask only “what is my best action?” Ask “what is my best action for each possible action or type of the opponent?”

21.2 Simultaneous Games With Complete Information

A simultaneous game is one where players choose without observing the other player's current move. Complete information means everyone knows the players, actions, and payoff table. There is still strategic uncertainty about what the opponent will choose, but there is no uncertainty about the rules of the game.

Definition 21.3 (Nash Equilibrium). *A Nash equilibrium is a strategy profile in which every player is playing a best response to the other players' strategies. Equivalently, no player can improve their utility by changing their own strategy while the other strategies are held fixed.*

Checking the four pure outcomes in the run/pass game gives:

Outcome	Offense wants to move?	Defense wants to move?
(Run, Run defense)	Yes, to Pass	No
(Run, Pass defense)	No	Yes, to Run defense
(Pass, Run defense)	No	Yes, to Pass defense
(Pass, Pass defense)	Yes, to Run	No

No pure outcome is stable. Every cell gives at least one side a profitable deviation.

The equilibrium is mixed. Let

$$p = \mathbb{P}(\text{offense runs}), \quad q = \mathbb{P}(\text{defense plays run defense}).$$

In a mixed equilibrium, each side chooses frequencies that make the other side indifferent between its pure actions. The game is still completely known, and we are just averaging across the known cells of the table.

If the defense uses run defense with frequency q , the offense's average conversion rate from always running is

$$40q + 70(1 - q) = 70 - 30q,$$

while always passing gives

$$70q + 40(1 - q) = 40 + 30q.$$

Indifference requires

$$70 - 30q = 40 + 30q \implies q = \frac{1}{2}.$$

If the defense uses run defense more often than $1/2$, passing is better. If the defense uses it less often than $1/2$, running is better.

If the offense runs with frequency p , the defense's average stop rate from always choosing run defense is

$$60p + 30(1 - p) = 30 + 30p,$$

while always choosing pass defense gives

$$30p + 60(1 - p) = 60 - 30p.$$

Indifference requires

$$30 + 30p = 60 - 30p \implies p = \frac{1}{2}.$$

If the offense runs more often than $1/2$, run defense is better. If the offense runs less often than $1/2$, pass defense is better.

Therefore the mixed-strategy Nash equilibrium is

$$\mathbb{P}(\text{Run}) = \frac{1}{2}, \quad \mathbb{P}(\text{Run defense}) = \frac{1}{2}.$$

The equilibrium conversion probability is

$$70 - 30 \left(\frac{1}{2} \right) = 55\%.$$

The point is not that coaches should randomize for its own sake. The point is that predictable behavior is exploitable. In equilibrium, the offense's run rate makes the defense indifferent between run defense and pass defense, and the defense's run-defense rate makes the offense indifferent between run and pass.

21.3 Sequential Games With Complete Information

A game can still have complete information even when the moves happen in order.

Consider a particular pre-snap sequence:

1. The offense chooses a formation: Heavy or Spread.
2. The defense observes the formation and chooses an alignment: Stack the box or Nickel shell.
3. The offense observes the alignment and can audible to Run or Pass.

The payoff table below gives the offense's conversion probability after the formation and defensive alignment are set:

Formation	Defense alignment	Final offensive call	
		Run	Pass
Heavy	Stack	45	55
Heavy	Nickel	75	30
Spread	Stack	35	70
Spread	Nickel	60	45

Because the offense moves last, backward induction solves the game.

Step 1: Offense's audible decision.

Formation	Defense alignment	Offense's best final call	
Heavy	Stack	Pass	(55 > 45)
Heavy	Nickel	Run	(75 > 30)
Spread	Stack	Pass	(70 > 35)
Spread	Nickel	Run	(60 > 45)

Step 2: Defense anticipates the audible.

Formation	Value if Stack	Value if Nickel	Defense choice
Heavy	55	75	Stack
Spread	70	60	Nickel

The defense minimizes the offense's conversion probability. Against Heavy, Stack limits the offense to 55. Against Spread, Nickel limits the offense to 60.

Step 3: Offense chooses the formation.

$$V(\text{Heavy}) = 55, \quad V(\text{Spread}) = 60.$$

The offense chooses Spread.

Definition 21.4 (Subgame-Perfect Equilibrium). *A subgame-perfect equilibrium is a strategy profile that is a Nash equilibrium after every possible history in a sequential game. In finite perfect-information games, we find it by backward induction.*

The subgame-perfect equilibrium is:

Initial offense strategy:	Choose Spread.
Defense strategy:	Stack after Heavy; Nickel after Spread.
Audible strategy:	Pass after Stack; Run after Nickel, for either formation.

The realized path is

$$\text{Spread} \longrightarrow \text{Nickel} \longrightarrow \text{Run},$$

with conversion probability 60%.

The important difference from the simultaneous game is that a strategy now means a full contingent plan. The defense's strategy must say what it would do after Heavy and after Spread, even though only one formation is actually chosen. The offense's audible strategy must say what it would do after each possible defensive alignment.

21.4 Incomplete Information and Bayesian Equilibrium

Complete information means everyone knows the payoff table. Incomplete information means at least one player has private information: a type, injury, blitz plan, coverage plan, personnel package, or constraint that the other player does not observe.

Incomplete-information games add beliefs about hidden possibilities. Expected utility is therefore taken over hidden types, not just over observed actions.

Definition 21.5 (Type). *A type is private information that changes a player's payoffs, available actions, or likely strategy. Other players may know the distribution of possible types without knowing the realized type.*

Definition 21.6 (Expected Utility). *Expected utility is the probability-weighted average utility over the hidden states or types a player thinks might be true.*

Definition 21.7 (Bayesian Equilibrium). *A Bayesian equilibrium specifies a strategy for each type of each player, plus beliefs about hidden types. Each strategy must maximize expected utility given those beliefs, and beliefs must be updated by Bayes' rule whenever the relevant information is observed.*

A Bayesian game adds one more step to rational choice. A player begins with a prior over hidden types t , observes a signal s , updates that prior to a posterior, and then chooses the action that performs best under that posterior:

$$\mathbb{P}(t \mid s) = \frac{\mathbb{P}(s \mid t)\mathbb{P}(t)}{\sum_{t'} \mathbb{P}(s \mid t')\mathbb{P}(t')}.$$

That posterior is what turns hidden information into a decision rule. If the player can choose an action a , the rational choice after seeing s is the one with the highest expected utility:

$$U(a \mid s) = \sum_t \mathbb{P}(t \mid s) u(a, t).$$

So Bayesian equilibrium means: beliefs are updated correctly from the prior, and each player's strategy is optimal given those posterior beliefs.

21.4.1 Example: Short-Yardage Scheme on 3rd-and-1

For a given game, suppose the defensive coordinator sets one short-yardage scheme: either *sell out to stop the run* or *play it balanced*. From scouting, the offense treats those two possibilities as equally likely:

$$\mathbb{P}(S) = \frac{1}{2}, \quad \mathbb{P}(B) = \frac{1}{2}.$$

The coordinator knows which scheme was installed for the game. The offense does not, but it does know this prior before seeing any new signal.

Suppose the pre-snap formation is either *Loaded box* or *Two-high*. From scouting and league-wide short-yardage data, the offense treats the scheme-contingent signal frequencies as known:

	Loaded box	Two-high
Sell-out (S)	$\frac{3}{4}$	$\frac{1}{4}$
Balanced (B)	$\frac{1}{4}$	$\frac{3}{4}$

A sell-out scheme usually looks heavier in the box, while a balanced scheme more often leaves a lighter shell, but neither signal is perfectly revealing.

The offense's conversion probabilities are:

	S	B
Run	20	80
Pass	80	40

Against a sell-out scheme, passing is much better than running. Running is better against a balanced scheme, but the gap is smaller. Given the defense's pre-snap formation, what should the offense run?

After a loaded box. Bayes' rule gives

$$\mathbb{P}(S \mid \text{Loaded}) = \frac{\frac{3}{4} \cdot \frac{1}{2}}{\frac{3}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2}} = \frac{3}{4}.$$

So the offense evaluates

$$\mathbb{E}(\text{Run} \mid \text{Loaded}) = 20 \left(\frac{3}{4} \right) + 80 \left(\frac{1}{4} \right) = 35,$$

$$\mathbb{E}(\text{Pass} \mid \text{Loaded}) = 80 \left(\frac{3}{4} \right) + 40 \left(\frac{1}{4} \right) = 70.$$

The best response is **Pass**.

After a two-high shell. Bayes' rule gives

$$\mathbb{P}(S \mid \text{Two-high}) = \frac{\frac{1}{4} \cdot \frac{1}{2}}{\frac{1}{4} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2}} = \frac{1}{4}.$$

So the offense evaluates

$$\mathbb{E}(\text{Run} \mid \text{Two-high}) = 20 \left(\frac{1}{4} \right) + 80 \left(\frac{3}{4} \right) = 65,$$

$$\mathbb{E}(\text{Pass} \mid \text{Two-high}) = 80 \left(\frac{1}{4} \right) + 40 \left(\frac{3}{4} \right) = 50.$$

The best response is **Run**.

The reduced Bayesian equilibrium is

Defense type S :	Loaded box with probability $3/4$
Defense type B :	Loaded box with probability $1/4$
Offense after Loaded box:	Pass
Offense after Two-high:	Run

with beliefs

$$\mathbb{P}(S \mid \text{Loaded}) = \frac{3}{4}, \quad \mathbb{P}(S \mid \text{Two-high}) = \frac{1}{4}.$$

The logic is the same as in the complete-information examples, but now the best response depends on a posterior belief about a hidden defensive scheme rather than on a directly observed defensive action.

21.5 Summary

- Utility turns an outcome into a decision objective.
- A best response is rational action against a specified opponent action or strategy.
- A simultaneous complete-information equilibrium is a set of mutual best responses.
- A sequential complete-information equilibrium is a set of mutual best responses after every possible history.
- A Bayesian equilibrium adds hidden types, expected utility, and Bayes' rule.

The common thread is self-consistency. A proposed strategy profile is an equilibrium only if every player would still choose their assigned strategy after thinking through what the other players are doing.